Mind World

A branching, scroll-scrubbed portfolio. Design & build specification.

`arpit2412.github.io` · `worktree` `~/Desktop/Resume/site-world` · `branch` `world/v2`
`Drafted 2026-07-10` · `Status:` **`blocked on credits only`** `(photos no longer required — see §8.1)`

1. The concept

The page opens on Arpit, full body, standing. The visitor scrolls; the camera pushes forward into his head. Inside is a **hub** — a chamber with doors. Each door is a domain of the work: Research, Engineering, Film & VFX, Writing, Honors. Choosing one flies the camera into that room, where the work lives. Scrolling back up surfaces you through the hub, where you can choose again.

It is not a linear tour. It is a space you re-enter. Every room is reachable from every other room, because every room returns you to the hub first, and the hub is one move from anywhere.

> **Branching costs less than linear, not more.** Scrubbing is bidirectional: scrolling backwards through a clip plays it in reverse. The *hub* → *Engineering* flight, reversed, *is* the *Engineering* → *hub* flight. Every return journey is free. Five rooms therefore need five edges, not ten. The branching graph costs **74 credits**; the linear eight-chapter version cost **91**.

2. Structure

```
           [ PORTRAIT ]  full body, locked off
                 |  scroll → camera pushes into the head
                 v
            [  HUB  ]  a chamber, five doors, ambient
             /  |  |  \  \
            /   |  |   \  \        each edge = ONE generated clip
           v    v  v    v  v       reverse-scrub gives the return free
      RESEARCH  |  |   WRITING  HONORS
             ENGINEERING
                FILM & VFX

  surfacing from any room = that room's edge, played backwards
  → any room is two moves from any other room, via the hub
```

Room contents (all existing content preserved)

ROOM	ANCHOR	HOLDS
Research	`#research`	Six research areas, selected publications
Engineering	`#work`	Project cards, the Lifecycle block (pretrain → unlearn)
Film & VFX	`#films`	Nine films, TikTok / AI Parameters content
Writing	`#writing`	Blog posts, news
Honors	`#honors`	Patents (US + UK), grant, awards
Journey	`#journey`	Career timeline — surfaces in the hub itself

3. Cost

All figures below were measured with live `get_cost` preflights against the Higgsfield MCP, not estimated.

ITEM	QTY	UNIT	CREDITS
Stills — `nano_banana_pro`, 2K, 16:9	7	2	14
Dive clips — `cinematic_studio_video_v2`, 5s, silent	7	5	35
Hub → room edges	5	5	25
Room → hub edges *(reverse-scrub)*	5	0	0
Base total			**74**
With realistic 2× rerolls			148

Model choice matters more than plan choice

VIDEO MODEL	PER 5S CLIP	BASE BUILD	FITS STARTER (270CR)?
`cinematic_studio_video_v2`	5.0	74	Yes, with 3× headroom
`seedance_2_0` fast, 720p	17.5	224	Base only, zero rerolls
`seedance_2_0` std	22.5	284	No — overshoots

> **Starter is a gamble on model access, not on credits.** Starter's own feature list says “access to *selected* models” and names only Seedance 2.0 Fast / Mini. If `cinematic_studio_video_v2` is not included, the build cannot complete on 270 credits. **First spend must be one 5-credit test clip on that exact model.** If it is refused, upgrade to Plus.

> **MCP and web prices disagree.** The MCP checkout offers Plus at **$49/mo (1,000 cr)** or **$29/mo annual**; higgsfield.ai advertises $59/mo for 1,200. Also: the MCP's own compliance note states *“unlimited models and free generations are accessible only via higgsfield.ai and are not accessible on MCP/CLI.”* All “7 unlimited models” bullets are worth nothing for this build — it runs through the MCP. Compare tiers on raw credits and model unlocks only.

4. Blockers

B1 — No credits. `list_workspaces` returns one workspace: `plan_type: "free"` , `credits: 0` . The MCP is connected and authenticated; it simply has no balance. Nothing generates until this is resolved. *Owner: Arpit.*

B2 — No photographs. The repo contains one image, `/arpit.png` . The concept requires Arpit's likeness to survive seven generated scenes. Two routes:

- `image_to_3d` → a real GLB mesh, orbitable, but needs `three.js` and reproduces only what is in the source photo.
- **Soul training** (5–20 photos, ~10 min) → a consistent face across every shot. Recommended — "inside my mind" fails if the face drifts between rooms.

Needed: full-body photos, several angles, good light. *Owner: Arpit.*

B3 — Film footage is not ours to publish. Furiosa, Deadpool, Mickey 17, Mortal Kombat II and the rest are owned by Warner Bros., Disney, Netflix et al. VFX work at Rising Sun Pictures is normally under NDA. Playing film clips or GIFs on a public portfolio is a live exposure, not a theoretical one.

The existing site already handles this correctly — `data/films.ts` uses Wikipedia poster art plus license-free Unsplash "mood" banners, and says so in a comment. **Keep that.** Arpit's own TikTok / AI Parameters content is his and can play freely. For the films: poster art plus a line on what shipped, or generated plates that evoke the work without being the work. *Owner: Arpit — decision required.*

5. Engineering

What survives

`scrubEngine.ts` — forked from scroll-world (MIT, oso95/scroll-world). It is segment-based and does not care what the chapters are, so the graph rewrite does not touch it. Two bugs were fixed against upstream:

- **Frame-rate-dependent smoothing.** Upstream runs `cur += (target − cur) * 0.18` once per rAF, which converges twice as fast at 120 Hz as at 60 Hz. Now solved for real frame delta.
- **Layout thrash.** Upstream measured element offsets inside the scroll handler. Offsets are now cached and recomputed on resize / ResizeObserver.

Kept as a lerp, never a spring: spring overshoot on a media seek scrubs past the target frame and visibly rewinds.

What must be rebuilt

- The chapter *chain* becomes a small *graph* with a router. Hub is a choice state; each room is its own scroll track; deep links (`#engineering`) drop straight in. This is real work, not a config change.
- Reverse-scrub as the return journey needs explicit handling at the hub boundary.

> **Opacity conflict.** On `origin/master`, `Research`, `Timeline` and `Writing` are `bg-surface` — fully opaque — and every card is `bg-card`. A fixed video stage behind them shows through in only five of nine sections. That alternating opacity *is* the current design's banding rhythm. Making the video visible everywhere is a deliberate redesign, not a bug fix.

> **Film grain does not compress.** Synthetic full-frame temporal noise at the pipeline's specified `crf 20` produced **273 MB** across 26 short placeholder clips. Generated footage behaves the same. Mitigate with low-amplitude grain in prompts, AV1/HEVC (grain synthesis) with an h264 fallback, or a CDN.

Hosting

`arpit2412.github.io` is a GitHub Pages hostname and **cannot** be pointed at Vercel. Moving hosts means a new domain. Nothing in the build assumes a host; the engine fetches clips as same-origin Blobs, so an absolute CDN URL needs CORS on the bucket.

6. Order of work

1. Arpit: supply photos; decide the films question; buy credits.
2. **2 credits** — one still of Arpit. He approves the face before anything else is spent.
3. **5 credits** — one test clip on `cinematic_studio_video_v2`. Proves model access *and* motion quality.
4. Remaining ~67 credits — the other six stills, then the dives, then the five hub edges.
5. Wire the graph router; rebuild the chapter map against the real components.
6. Encode desktop + mobile variants; decide repo vs CDN on measured bytes.

Stills are rerolled at 2 credits; clips at 5. Approve every still before generating any motion — it is the cheapest place to be wrong.

7. Record of corrections

Three mistakes were made and corrected. They are logged so they are not repeated.

- **Built against a stale checkout.** The local tree was 5 commits behind `origin/master`; the live site is a ground-up redesign. `git log` was run; `git fetch` was not. Components that were wired (`Publications.tsx`, `FilmsMarquee.tsx`) no longer exist upstream. The stale tree also carries a wrong grant figure — A$2.1M, corrected upstream to **A$1.2M**.
- **Designed without watching the reference.** The scroll-world demo is a reel of *three* sites — an isometric diorama *and* two photoreal ones (luxury real estate, automotive). The claim that “photoreal is high risk, the engine defaults to diorama” was drawn from a README, not the artifact, and was wrong.
- **Proposed abstraction over photorealism** on the strength of that error, producing a dark, grainy “latent space” theme that resembled none of the references.

8. Revision — the narrative spine

The v1 structure was five résumé folders inside a cinematic wrapper. Doors labelled *Research / Engineering / Film / Writing / Honors* are navigation, not story. Adopted: the five worlds are five expressions of one process.

DISPLAY VERB	WORLD	ANCHOR (UNCHANGED)	NARRATIVE ROLE
QUESTION	Research	`#research`	Something unresolved must be understood
BUILD	Engineering	`#work`	Research survives contact with real systems
CREATE	Film & VFX	`#films`	The technology disappears; the experience remains
EXPLAIN	Writing	`#writing`	Understanding is incomplete until it is explained
IMPACT	Impact *(was Honors)*	`#impact`	Ideas leave evidence

Verbs are art direction, not information architecture. Anchors, `<h2>` s and nav labels stay the nouns. Recruiters Ctrl-F “Publications”; search engines index headings. The verb sits above the noun, never instead of it.

8.1 The opening is now a silhouette — this unblocks B2

Do not fly into a literal skull. The figure dissolves into particles / neural traces, revealing the hub. Consequences, all good:

- A silhouette needs **no likeness** → **Soul training is no longer required**. Blocker B2 downgrades from blocking to optional. Arpit's face no longer has to survive seven generated scenes without drifting.
- Removes the uncanny-valley risk of a generated full-body figure.
- `ParticleField.tsx` **already exists** in this repo — a live canvas node/edge network in the Hero. The silhouette → particles transition renders **in-browser: zero credits, no video seam**.
- The opening portrait still + dive may not need generating at all. Cost drops below 74 cr.

8.2 Returns get narrative lines (free)

The reverse-scrub is an implementation detail the visitor never perceives — they simply scroll back. Overlaying a different line on the way up costs **zero credits**, because the text is DOM, not video.

- Leaving Research → *“An idea has been discovered. What should it become?”*
- Leaving Engineering → *“A system has been built. Where can it create impact?”*
- Leaving Film → *“Technology became an experience. How can it be understood?”*
- Leaving Writing → *“The idea has been explained. What evidence did it leave?”*

8.3 Rejected / unresolved

Rejected — gating the ending. The proposal shows “Build What's Next” only after two or more worlds are visited. That hides the conversion point from the majority, who open one room and leave, and it needs visit-state tracking. **The CTA is always present, quieter at first, growing in emphasis. Never absent.**

> **Unresolved — the hub cannot be symmetric and hierarchical.** The proposal wants five doors, then ranks them (Research + Engineering primary; Film + Writing differentiators; Impact as proof). Five equal doors *say* the worlds are equal. Either the doors differ physically — size, light, camera proximity — or symmetry is accepted. **Must be decided before the hub still is generated; it is baked into that image.** *Owner: Arpit.*

> **Unresolved — the timeline cannot be a "subtle orbit."** `Timeline.tsx` is the tallest component on the site and drives its own scroll-linked spine. It cannot be ornamental decoration *and* hold the full career (DRDO/WESEE → Rising Sun Pictures → TikTok → AIML/CSIRO). Either a real sixth destination, or the orbit shows five org names and links out. *Owner: Arpit.*

8.4 Copy

Opening: **ARPIT GARG** — *I study intelligence, build it into systems, and bring it into the real world.*
Hub: **ONE MIND. FIVE EXPRESSIONS.**
Closing: **THE NEXT WORLD DOES NOT EXIST YET. Let's build it.**

8.5 Cost impact

Adding the sixth central destination ("Build What's Next") costs ~12 cr (1 still + 1 dive + 1 edge). Removing the generated opening portrait + dive in favour of the in-browser silhouette saves ~7 cr. Net build stays under **~80 credits**, still inside Starter's 270 with rerolls — subject to the `cinematic_studio_video_v2` access test.

9. Decisions taken

Arpit delegated the two open calls with the direction: *make it premium, don't worry about credits.*

9.1 The hub is ranked, not symmetric

Five equal doors read as a menu. Hierarchy is carried by **composition, not by labels** — it is baked into the hub still, which is why this had to be settled first.

- **QUESTION** (Research) and **BUILD** (Engineering) — two large apertures flanking the core, near, at eye level, warm key light. These are the primary identity.
- **CREATE** (Film) and **EXPLAIN** (Writing) — offset, further back, cooler, smaller. Differentiators.
- **IMPACT** — set behind and below the core, the deepest aperture. It is evidence; you reach it last and it feels earned.

9.2 The career timeline is the descent, not a door

> **This resolves both open conflicts at once.** `Timeline.tsx` is the tallest component on the site and drives its own scroll-linked spine. It cannot be an ornamental orbit. But it should not be a sixth door either — the career is not one of the five expressions, it is *how the mind was built*.
>
> So the timeline **is the corridor**. After the silhouette dissolves, the camera descends through the career — DRDO/WESEE → Rising Sun Pictures → TikTok → AIML/CSIRO — and arrives at the hub. The descent already needs scroll length; a tall scroll-driven spine is exactly what belongs there.
>
> Narratively it earns its place: *you did not jump between industries at random; you moved repeatedly toward harder, higher-impact intelligent systems*. The visitor understands that **before** they choose a door.

9.3 “Build What's Next” is the core

Not a hidden sixth destination, and never gated behind exploring two worlds. The glowing core at the centre of the hub **is** the contact path. It is the most prominent object in the chamber from the first second — it simply does not announce itself as a call to action until the visitor has been somewhere. Prominence grows; presence is constant. The one-room visitor still sees it.

9.4 Premium production settings

ASSET	SETTING	UNIT	QTY	CREDITS
Silhouette opening	in-browser `ParticleField`	0	1	0
Stills	`nano_banana_pro` , **4K**, 16:9	4	8	32
Dive clips	`cinematic_studio_video_v2` , **pro**, **8s**, silent	12	8	96
Hub → room edges	same, `start_image` + `end_image`	12	5	60
Room → hub edges	reverse-scrub	0	5	0
Premium base				**188**
With 2× rerolls				376

8-second clips over 5-second ones buy longer scroll travel per room and a slower, less frantic camera — the single biggest lever on whether this reads as premium or as a gimmick. `pro` mode and 4K stills are the rest.

> **Plan decision: Starter is now ruled out.** Premium base is 188 cr — it fits inside 270, but 2× rerolls (376 cr) do not, and connectors are precisely the shots that need rerolling. **Plus (1,200 cr) is the floor**; via the MCP checkout that is $49/mo, or $29/mo billed annually. Ultra (3,000 cr) buys 3× reroll headroom. Starter would also still carry the unresolved risk that `cinematic_studio_video_v2` is not among its "selected models".

> **Still blocked. The account holds 0 credits.** Re-checked at time of writing: `{"credits": 0, "subscription_plan_type": "free"}`. Every figure above is a measured `get_cost` preflight, which costs nothing and submits no job. Not a single asset can be generated until a plan is purchased.
>
> Photographs are **no longer required** — the silhouette opening removed that blocker (§8.1).

9.5 First spend, in order

1. **4 cr** — the hub still, at 4K. It encodes the ranked composition. Approve it before anything else exists.
2. **12 cr** — one `pro` 8s clip from that still. Proves model access *and* that the camera reads as cinematic rather than as an AI pan.
3. Remaining ~172 cr — the other seven stills, then the dives, then the five edges.

Stills reroll at 4 cr, clips at 12. Every still is approved before any motion is generated from it.
